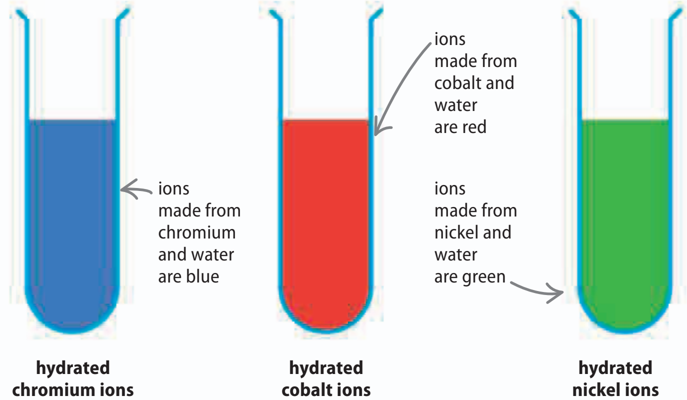
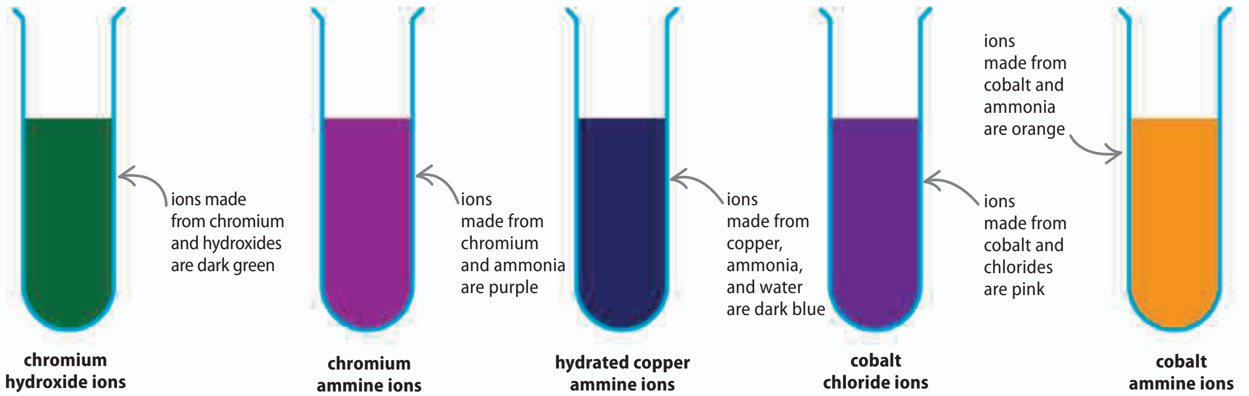
Complex colors

A complex ion has a metal ion at its center with a number of other molecules or ions surrounding it. Transition elements form huge complex ions with molecules such as water, ammonia, and chlorine. The structures of these ions are very complicated and vary enormously. The wavelengths of light that these ions absorb and emit also varies enormously, so the compounds that they form come in a rainbow of different colors.





Rare earth metals

The 30 rare earth metals are normally shown at the bottom of the periodic table, as there is no room to place them between Groups 2 and 3. They form in a similar way to the transition metals. Large atoms, from the sixth period on, grow by back-filling electrons, although this time the electrons are added two shells down, not one. The fourth and fifth atomic shells have room for 32 electrons.

▽ **Huge atoms**
Lanthanides are used to make high-tech alloys, while all of the actinides are radioactive. Uranium and thorium are used as nuclear fuels.

La LANTHANUM	Ce CERIUM	Pr PRASEODYMIUM	Nd NEODYMIUM	Pm PROMETHIUM	Sm SAMARIUM	Eu EUROPIUM	Gd GADOLINIUM	Tb TERBIUM	Dy DYSPROSIUM	Ho HOLMIUM	Er ERBIUM	Tm THULIUM	Yb YTTERBIUM	Lu LUTETIUM
Ac ACTINIUM	Th THORIUM	Pa PROTACTINIUM	U URANIUM	Np NEPTUNIUM	Pu PLUTONIUM	Am AMERICIUM	Cm CURIUM	Bk BERKELIUM	Cf CALIFORNIUM	Es EINSTEINIUM	Fm FERMIUM	Md MENDELEVIUM	No NOBELIUM	Lr LAWRENCIUM

the rare earth metals are also referred to as the lanthanides and actinides, accordng to the first element in each period

many of the radioactive actinides only exist for a few seconds in laboratories